

DBSE WEEKLY LAB EXPERIMENTS

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SECTION – 7

CSE

DDL, DML COMMANDS

The screenshot shows the SQL Developer interface with a script editor containing the following SQL commands:

```
1 CREATE DATABASE IF NOT EXISTS practice_db;
2 USE practice_db;
3
4 DROP TABLE IF EXISTS bank_backup;
5 DROP TABLE IF EXISTS customer_transactions;
6 DROP TABLE IF EXISTS bank_transactions;
7
8 CREATE TABLE bank_transactions (
9     txn_id INT PRIMARY KEY,
10    customer_name VARCHAR(50),
11    branch_name VARCHAR(50),
12    transaction_type VARCHAR(20),
13    amount DECIMAL(10,2),
14    transaction_date DATE
15 );
16
17 ALTER TABLE bank_transactions
18 ADD account_no VARCHAR(20);
19
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	09:11:33	CREATE DATABASE IF NOT EXISTS practice_db	1 row(s) affected	0.016 sec
2	09:11:33	USE practice_db	0 row(s) affected	0.000 sec
3	09:11:33	DROP TABLE IF EXISTS bank_backup	0 row(s) affected, 1 warning(s): 1051 Unknown table 'practice_db.bank_backup'	0.000 sec
4	09:11:33	DROP TABLE IF EXISTS customer_transactions	0 row(s) affected, 1 warning(s): 1051 Unknown table 'practice_db.customer_transactions'	0.000 sec
5	09:11:33	DROP TABLE IF EXISTS bank_transactions	0 row(s) affected, 1 warning(s): 1051 Unknown table 'practice_db.bank_transactions'	0.000 sec
6	09:11:33	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	0 row(s) affected	0.047 sec
7	09:11:33	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec
8	09:11:33	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec

The screenshot shows the SQL Developer interface with a script editor containing the following SQL commands:

```
26 INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date, account_no) VALUES
27 (101, 'Kairav', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
28 (102, 'Kairav', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
29 (103, 'Kairav', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
30 (104, 'Kairav', 'Vizag', 'Deposit', 8000, '2024-01-10'),
31 (105, 'Kairav', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
32 (106, 'Kairav', 'Vizag', 'Deposit', 15000, '2024-01-12'),
33 (107, 'Kairav', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
34 (108, 'Kairav', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
35 (109, 'Kairav', 'Vijayawada', 'Withdrawal', 4000, '2024-01-15');
```

The Result Grid shows the data inserted into the table:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Kairav	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Kairav	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Kairav	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Kairav	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Kairav	Hyderabad	Withdrawal	3500.00	2024-01-11	NULL
106	Kairav	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Kairav	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
4	09:15:26	DROP TABLE IF EXISTS customer_transactions	0 row(s) affected, 1 warning(s): 1051 Unknown table 'practice_db.customer_transactions'	0.000 sec
5	09:15:26	DROP TABLE IF EXISTS bank_transactions	0 row(s) affected, 1 warning(s): 1051 Unknown table 'practice_db.bank_transactions'	0.015 sec
6	09:15:26	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	0 row(s) affected	0.047 sec

Navigator: exp2 SQL File 3* x

SCHEMAS

Filter objects

sys

Administration Schemas

Information

No object selected

```
38 INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
39 VALUES (111, 'Kairav', 'Vizag', 'Deposit', 7000, '2024-01-18');
40
41 UPDATE bank_transactions
42 SET amount = 5000
43 WHERE txn_id = 105;
44
45 DELETE FROM bank_transactions
46 WHERE txn_id = 111;
```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Kairav	Hyderabad	Deposit	5000.00	2024-01-05	ROLL
102	Kairav	Hyderabad	Withdrawal	2000.00	2024-01-06	ROLL
103	Kairav	Vijayawada	Deposit	12000.00	2024-01-08	ROLL
104	Kairav	Vizag	Deposit	8000.00	2024-01-10	ROLL
105	Kairav	Hyderabad	Withdrawal	5000.00	2024-01-11	ROLL
106	Kairav	Vizag	Deposit	15000.00	2024-01-12	ROLL
107	Kairav	Vijayawada	Withdrawal	1000.00	2024-01-13	ROLL

bank_transactions 7 x bank_transactions 8 bank_transactions 9

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	09:15:26	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	0 row(s) affected	0.047 sec
7	09:15:26	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
8	09:15:26	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
9	09:15:26	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected	0.000 sec
10	09:15:26	RENAME TABLE customer_transactions TO bank_transactions	0 row(s) affected	0.000 sec
11	09:15:26	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transac...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
12	09:15:26	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transac...	1 row(s) affected	0.000 sec
13	09:15:26	UPDATE bank_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec

Object Info Session

Navigator: exp2 SQL File 3* x

SCHEMAS

Filter objects

sys

Administration Schemas

Information

No object selected

```
44
45 DELETE FROM bank_transactions
46 WHERE txn_id = 111;
47
48 SELECT * FROM bank_transactions;
49
50 SELECT *
51 FROM bank_transactions
52 WHERE transaction_type='Deposit';
```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Kairav	Hyderabad	Deposit	5000.00	2024-01-05	ROLL
102	Kairav	Hyderabad	Withdrawal	2000.00	2024-01-06	ROLL
103	Kairav	Vijayawada	Deposit	12000.00	2024-01-08	ROLL
104	Kairav	Vizag	Deposit	8000.00	2024-01-10	ROLL
105	Kairav	Hyderabad	Withdrawal	5000.00	2024-01-11	ROLL
106	Kairav	Vizag	Deposit	15000.00	2024-01-12	ROLL
107	Kairav	Vijayawada	Withdrawal	1000.00	2024-01-13	ROLL

bank_transactions 7 x bank_transactions 8 bank_transactions 9

Output

Action Output

#	Time	Action	Message	Duration / Fetch
14	09:15:26	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected	0.000 sec
15	09:15:26	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
16	09:15:26	SELECT * FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec

Navigator

SCHMAS

Filter objects

sys

Administration Schemas

Information

No object selected

exp2 SQL File 3

Limit to 1000 rows

```
54 SELECT *
55 FROM bank_transactions
56 ORDER BY amount DESC;
57
58 START TRANSACTION;
59 UPDATE bank_transactions
60 SET amount = 6000
61 WHERE txn_id = 101;
62 COMMIT;
```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Karav	Hyderabad	Deposit	5000.00	2024-01-05	0000
102	Karav	Hyderabad	Withdrawal	2000.00	2024-01-06	0000
103	Karav	Vijayawada	Deposit	12000.00	2024-01-08	0000
104	Karav	Vizag	Deposit	8000.00	2024-01-10	0000
105	Karav	Hyderabad	Withdrawal	5000.00	2024-01-11	0000
106	Karav	Vizag	Deposit	15000.00	2024-01-12	0000
107	Karav	Vijayawada	Withdrawal	1000.00	2024-01-13	0000

bank_transactions 7 x bank_transactions 8 bank_transactions 9

Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
17	09:15:26	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
18	09:15:26	START TRANSACTION	0 row(s) affected	0.000 sec
19	09:15:26	UPDATE bank_transactions SET amount = 6000 WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
20	09:15:26	COMMIT	0 row(s) affected	0.000 sec

Navigator

SCHMAS

Filter objects

sys

Administration Schemas

Information

No object selected

exp2 SQL File 3

Limit to 1000 rows

```
67 UPDATE bank_transactions
68 SET amount = 99999
69 WHERE txn_id = 102;
70
71 ROLLBACK TO Before_Update;
72 COMMIT;
73
74 START TRANSACTION;
75 UPDATE bank_transactions
76 SET amount = 7000
```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Karav	Hyderabad	Deposit	5000.00	2024-01-05	0000
103	Karav	Vijayawada	Deposit	12000.00	2024-01-08	0000
104	Karav	Vizag	Deposit	8000.00	2024-01-10	0000
106	Karav	Vizag	Deposit	15000.00	2024-01-12	0000
108	Karav	Hyderabad	Deposit	9000.00	2024-01-14	0000
110	Karav	Vijayawada	Deposit	11000.00	2024-01-16	0000

bank_transactions 7 bank_transactions 8 x bank_transactions 9

Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
23	09:15:26	UPDATE bank_transactions SET amount = 99999 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
24	09:15:26	ROLLBACK TO Before_Update	0 row(s) affected	0.016 sec
25	09:15:26	COMMIT	0 row(s) affected	0.000 sec
26	09:15:26	START TRANSACTION	0 row(s) affected	0.000 sec
27	09:15:26	UPDATE bank_transactions SET amount = 7000 WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
28	09:15:26	SAVEPOINT SP1	0 row(s) affected	0.000 sec
29	09:15:26	UPDATE bank_transactions SET amount = 9000 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
30	09:15:26	ROLLBACK TO SP1	0 row(s) affected	0.000 sec

Object Info Session

exp2 SQL File 3'

```
77 WHERE txn_id = 101;
78
79 • SAVEPOINT SP1;
80
81 • UPDATE bank_transactions
82   SET amount = 9000
83   WHERE txn_id = 102;
84
85 • ROLLBACK TO SP1;
86 • COMMIT;
87
```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Karav	Vizag	Deposit	15000.00	2024-01-12	1001
103	Karav	Vijayawada	Deposit	12000.00	2024-01-08	1002
110	Karav	Vijayawada	Deposit	11000.00	2024-01-16	1003
108	Karav	Hyderabad	Deposit	9000.00	2024-01-14	1004
104	Karav	Vizag	Deposit	8000.00	2024-01-10	1005

bank_transactions 7 bank_transactions 8 bank_transactions 9 x

Output

#	Time	Action	Message	Duration / Fetch
25	09:15:26	COMMIT	0 row(s) affected	0.000 sec
26	09:15:26	START TRANSACTION	0 row(s) affected	0.000 sec
27	09:15:26	UPDATE bank_transactions SET amount = 7000 WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
28	09:15:26	SAVEPOINT SP1	0 row(s) affected	0.000 sec
29	09:15:26	UPDATE bank_transactions SET amount = 9000 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
30	09:15:26	ROLLBACK TO SP1	0 row(s) affected	0.000 sec
31	09:15:26	COMMIT	0 row(s) affected	0.015 sec
32	09:15:26	CREATE TABLE bank_backup (id INT)	0 row(s) affected	0.016 sec

exp2 SQL File 3'

```
83 WHERE txn_id = 102;
84
85 • ROLLBACK TO SP1;
86 • COMMIT;
87
88 • CREATE TABLE bank_backup (
89   id INT
90 );
91
92 • TRUNCATE TABLE bank_backup;
93 • DROP TABLE bank_backup;
```

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Karav	Vizag	Deposit	15000.00	2024-01-12	1001
103	Karav	Vijayawada	Deposit	12000.00	2024-01-08	1002
110	Karav	Vijayawada	Deposit	11000.00	2024-01-16	1003
108	Karav	Hyderabad	Deposit	9000.00	2024-01-14	1004
104	Karav	Vizag	Deposit	8000.00	2024-01-10	1005

bank_transactions 7 bank_transactions 8 bank_transactions 9 x

Output

#	Time	Action	Message	Duration / Fetch
26	09:15:26	START TRANSACTION	0 row(s) affected	0.000 sec
27	09:15:26	UPDATE bank_transactions SET amount = 7000 WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
28	09:15:26	SAVEPOINT SP1	0 row(s) affected	0.000 sec
29	09:15:26	UPDATE bank_transactions SET amount = 9000 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
30	09:15:26	ROLLBACK TO SP1	0 row(s) affected	0.000 sec
31	09:15:26	COMMIT	0 row(s) affected	0.015 sec
32	09:15:26	CREATE TABLE bank_backup (id INT)	0 row(s) affected	0.016 sec
33	09:15:26	TRUNCATE TABLE bank_backup	0 row(s) affected	0.015 sec